SIMATS ENGINEERING

CSA16 – DATA WAREHOUSING AND DATA MINING

CO2 Assessment Tool 1 – Detailed Solutions

Course Outcome CO2 focuses on data preprocessing: data cleaning, integration, transformation and reduction. The following answers show the method, calculations and interpretation for all five questions in the provided assessment.

1. Covariance Analysis in Health Insurance (10 Marks)

Given data: Premium increase x = 3.2, 4.0, 5.5, 4.8; claims per 100 policyholders y = 15, 18, 25, 20.

Step 1: Calculate the means

x■ = Σx/n = (3.2 + 4.0 + 5.5 + 4.8)/4 = 17.5/4 = **4.375%**.

■ = Σy/n = (15 + 18 + 25 + 20)/4 = 78/4 = **19.5 claims**.

Step 2: Calculate deviations and their products

i	x■	y■	x■ – x■	y■ – ■	Product of deviations
1	3.2	15	-1.175	-4.5	5.2875
2	4.0	18	-0.375	-1.5	0.5625
3	5.5	25	1.125	5.5	6.1875
4	4.8	20	0.425	0.5	0.2125
Σ					12.2500

Step 3: Apply covariance formula

Population covariance: Cov(X,Y) = Σ[(x■ – x■)(y■ – ■)] / n = 12.25/4 = **3.0625**.

Interpretation: Covariance is positive, so higher premium increases tend to occur with higher claim frequency in this sample. Covariance indicates direction, not the strength in a standardized scale.

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

Purpose: Equal-frequency (equi-depth) binning sorts values and divides them into bins containing the same number of observations. Smoothing replaces noisy values with a representative value. The question does not specify bin count or smoothing method, so this solution assumes 3 bins of 9 values each and uses bin means.

Bin 1 (n = 9): 110, 112, 115, 115, 118, 120, 120, 122, 124

Mean = (110 + 112 + 115 + 115 + 118 + 120 + 120 + 122 + 124) / 9 = 1056 / 9 = **117.33**. After smoothing by bin mean, each of the 9 values in this bin is replaced by 117.33.

Bin 2 (n = 9): 124, 126, 126, 126, 126, 130, 135, 135, 138

Mean = (124 + 126 + 126 + 126 + 126 + 130 + 135 + 135 + 138) / 9 = 1166 / 9 = **129.56**. After smoothing by bin mean, each of the 9 values in this bin is replaced by 129.56.

Bin 3 (n = 9): 138, 138, 138, 140, 145, 150, 152, 160, 180

Mean = (138 + 138 + 138 + 140 + 145 + 150 + 152 + 160 + 180) / 9 = 1341 / 9 = **149.00**. After smoothing by bin mean, each of the 9 values in this bin is replaced by 149.00.

Smoothed data: 110–124 range → 121.78 repeated 9 times; second bin → 129.56 repeated 9 times; third bin → 146.78 repeated 9 times. These representative values reduce small fluctuations while preserving the general pattern.

3. Equal-Width Binning for Risk Categorization (5 Marks)

Given: 18 sorted BMI values, from minimum 18.5 to maximum 39.5. Equal-width binning divides the range into bins of identical width.

Range = max – min = 39.5 – 18.5 = 21.0. Assuming 3 bins, width = 21.0/3 = **7.0 BMI units**.

Use bins [18.5, 25.5], (25.5, 32.5], (32.5, 39.5]. The prompt says to replace values by the closest bin boundaries, so each value is replaced by whichever endpoint is closer.

Bin 1: 18.5, 19.2, 21.8, 24.9, 25.5 → 18.5, 18.5, 18.5, 25.5, 25.5.

Bin 2: 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4 → 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5.

Bin 3: 33.1, 33.6, 34.7, 38.2, 39.5 → 32.5, 32.5, 32.5, 39.5, 39.5.

Interpretation: Equal-width bins make the numeric range easier to categorize. Boundary smoothing reduces detail, so original BMI values should be retained when individual clinical decisions are required.

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

Purpose: Bin-median smoothing sorts values, groups them, calculates each group's median, and represents every value in a group by that median. The question does not specify bin count; assume 3 equal-frequency bins of 9 observations each.

Step 1: Sort the costs

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300.

Bin 1: 120, 140, 145, 145, 148, 150, 150, 150, 155. Since each bin has 9 values, the median is the 5th value = ■**148**.

Bin 2: 155, 155, 155, 160, 160, 165, 170, 170, 175. Since each bin has 9 values, the median is the 5th value = ■**160**.

Bin 3: 175, 175, 175, 180, 190, 200, 210, 230, 300. Since each bin has 9 values, the median is the 5th value = ■**190**.

Step 2: Reduced data

Replace all values in Bin 1 with ■148, all values in Bin 2 with ■160, and all values in Bin 3 with ■190. Thus the reduced representation is ■148 repeated 9 times, ■160 repeated 9 times and ■190 repeated 9 times.

Interpretation: The median represents the typical cost in each bin and is less affected by extreme values such as ■300 than a mean would be.

5. Standardization of Patient Age Data (5 Marks)

Given ages: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28.

(a) Mean absolute deviation (MAD)

Mean age = Σage/n = 330/10 = **33 years**.

Absolute deviations from 33 are: 15, 11, 8, 9, 5, 10, 0, 2, 23, 5. Their sum = 88.

MAD = Σ|x – x■|/n = 88/10 = **8.8 years**.

(b) Z-scores for the first four observations

Use population standard deviation: σ = √[Σ(x – x■)²/n] = √117.4 = **10.835** years.

Z-score formula: z = (x – x■)/σ.

For age 18: z = (18 – 33)/10.835 = **-1.38**.

For age 22: z = (22 – 33)/10.835 = **-1.02**.

For age 25: z = (25 – 33)/10.835 = **-0.74**.

For age 42: z = (42 – 33)/10.835 = **0.83**.

Conclusion: The MAD describes the average absolute distance from the mean age. Z-scores show how many standard deviations each observation is above or below the mean.

Important assumptions

Questions 2–4 do not state the number of bins. The detailed answers therefore explicitly assume 3 bins. Question 2 uses bin-mean smoothing; Question 3 uses 3 equal-width bins; Question 4 uses 3 equal-frequency bins. If your instructor specified a different bin count or boundary convention, the corresponding binning answers will change.

Student Name: _____ Register No.: _____